FORWARD DEPLOYED / PROPERTY / SAFETY CRITICAL / CASE STUDY

MAINTENANCE TRIAGE

CASE STUDY

10

A synthetic property manager tests triage that escalates safety reports first without delay.

PROPOSED RESEARCH / NOT YET VERIFIED

THE BUSINESS CONTEXT

A synthetic property manager looks after about 480 units and receives repair reports by email and phone and an app. Staff read each one and judge urgency under pressure. A gas smell or a flood buried in a long queue is the risk that matters most. Routine jobs also need grouping and routing.

WHO IS INVOLVED AND WHAT THEY NEED

01	**Property manager**	Needs every safety report in front of a person at once and routine work organised.
02	**Maintenance coordinator**	Wants duplicates merged and the right contractor and unit identified quickly.
03	**Safeguarding lead**	Needs concerns about people to reach the right person immediately and privately.

SCOPE AND HONEST LIMITS

- ⚠ Reports and units are synthetic so real tenant language is not fully represented.
- ⚠ No safety rating or certification is claimed and none would follow from testing.
- ⚠ Live use would need legal and regulatory review and tenant privacy controls.
- ⚠ Voice transcription quality is assumed rather than measured.

CURRENT WORKFLOW AND FAILURE POINTS

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PROPOSED RESEARCH / NOT YET VERIFIED

HOW THE WORK RUNS TODAY

1 TODAY

Reports arrive

Emails and app messages land in shared inboxes while phone calls are logged by hand.

2 TODAY

Read and judge

A coordinator reads each report and decides urgency from memory and instinct.

3 TODAY

Find the unit

The unit and tenant are looked up in a spreadsheet or the property system.

4 TODAY

Spot duplicates

Repeat reports about one leak are noticed only if the same person sees both.

5 TODAY

Pick a contractor

Contractors are chosen by habit and called or emailed one at a time.

6 TODAY

Update the tenant

Status is passed back by phone or email and the record is updated afterwards.

WHERE IT BREAKS

Buried emergencies	A gas, fire, flood or safeguarding report can sit unread among routine requests.
Inconsistent urgency	Different staff rate the same issue differently depending on workload.
Duplicate noise	Several reports of one fault create several jobs and contractor calls.
Weak records	Phone details and decisions are rarely captured in a way that can be reviewed.

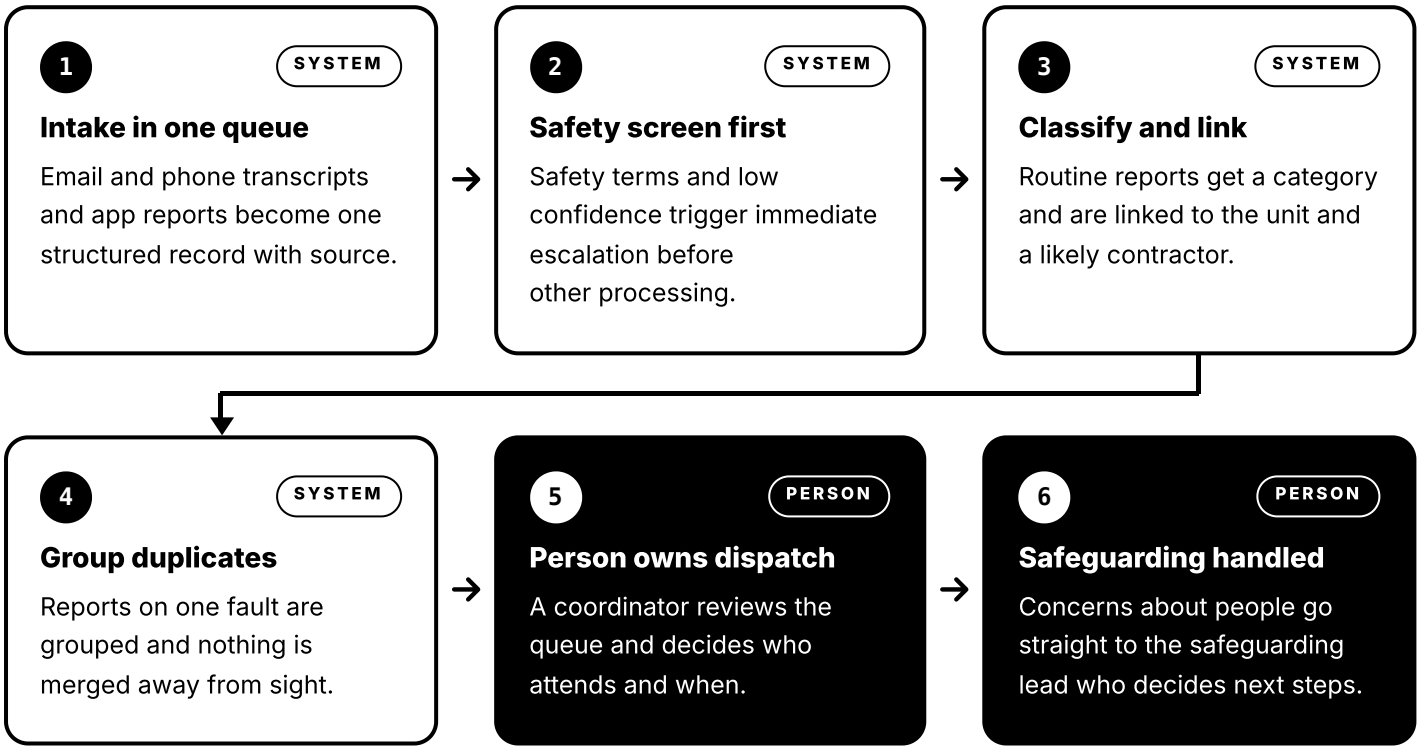
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REDESIGNED WORKFLOW AND DECISION AUTHORITY

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PROPOSED RESEARCH / NOT YET VERIFIED

THE REDESIGNED WORKFLOW



SYSTEM STEP PERSON DECISION

WHERE AUTHORITY STAYS WITH A PERSON

Who is dispatched and when	MAINTENANCE COORDINATOR	Dispatch carries cost and access and safety consequences that need a person.
Whether an escalation is closed	PROPERTY MANAGER	Only a person can judge that a safety risk is resolved.
How a safeguarding concern is handled	SAFEGUARDING LEAD	It involves people and legal duties that no system should decide.

ARCHITECTURE, DATA AND CONTROLS

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PROPOSED RESEARCH / NOT YET VERIFIED

SYSTEM ARCHITECTURE

1	**Intake adapters**	Normalise email and phone transcript and app reports into one record with timestamps.
2	**Safety gate**	Runs first with rules and a model and escalates on any safety term or doubt.
3	**Classification**	Assigns category and urgency to routine reports with a confidence score.
4	**Linking and grouping**	Ties reports to units and tenants and contractors and groups likely duplicates.
5	**Dispatch workspace**	Queue showing escalations first with a full log of every person decision.

CORE DATA ENTITIES

- **Report**
- **Unit**
- **Tenant**
- **Contractor**
- **Escalation**
- **Dispatch decision**

RISKS AND THE CONTROL FOR EACH

Safety report missed →	The safety gate runs before all else and any doubt or failure escalates to a person.
Model outage or timeout →	If classification fails the report is treated as urgent and sent to a person.
Duplicate hides a new hazard →	Grouped reports stay visible and any new safety term in a group triggers escalation.
Automatic dispatch →	The system never books a contractor; a person confirms every dispatch decision.

EVALUATION AND DELIVERY PLAN

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PROPOSED RESEARCH / NOT YET VERIFIED

HOW SUCCESS IS MEASURED

Safety recall	Share of labelled synthetic safety reports escalated; any miss is a failed test.
Time to escalation	Elapsed time from report receipt to a person seeing a safety escalation.
False escalation rate	Share of escalations a person judges routine after review.
Duplicate accuracy	Share of duplicate reports grouped correctly without wrong merges.
Routing correctness	Share of routine reports linked to the right unit and contractor type.

TEST PLAN

1. **Replay reports describing a gas smell in vague and garbled wording and confirm all escalate.**
2. **Force the classifier to fail and confirm every report is escalated and none are dropped.**
3. **Send a new hazard into a group of leak reports and confirm it escalates at once.**
4. **Send a safeguarding concern by phone transcript and confirm it reaches the lead only.**

DELIVERY PLAN

01 ALIGN	02 UNDERSTAND	03 DESIGN	04 DELIVER	05 LAUNCH
Agree escalation categories and who is on call for each.	Review past reports and how phone calls are recorded.	Design the safety gate and queue and the failure behaviour.	Build the pipeline and run safety recall tests on synthetic data.	Run in shadow beside staff before any live reliance.
DECISION Which reports count as immediate	**DECISION** What the safe default is when unsure	**DECISION** How much over escalation is tolerable	**DECISION** Whether any missed case blocks release	**DECISION** Whether to move to supervised live use

DELIVERABLES

- Synthetic report and unit dataset
- Escalation rule catalogue
- Safety recall test suite
- Dispatch queue design
- Evaluation report with failure cases